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Вар 47

$$y_t'' + 9y_t' + 20y_t = 5 \quad y_0 = 4 \quad y_0' = 5$$

$$y_t' = z_t \quad y_t'' = z_t'$$

$$\begin{cases} y_t' = z_t \\ z_t' = -20y_t + 9z_t + 5 \end{cases}$$

$$X(t) = \begin{bmatrix} y_t \\ z_t \end{bmatrix} \quad X_0 = \begin{bmatrix} y_0 \\ z_0 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} = d$$

$$A = \begin{bmatrix} 0 & 1 \\ -20 & 9 \end{bmatrix}; \quad b = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

$$x(t) = Ax_t + b; \quad x_0 = d; \quad x_t = e^{At}x_0 + \int_0^t e^{A(t-s)}b \, ds$$

$$\det(A) \neq 0$$

$$x_t = e^{At}x_0 + A^{-1}(e^{At} - E)b$$

$$f(A) = f(\lambda_1) \frac{A - \lambda_2 E}{\lambda_1 - \lambda_2} + f(\lambda_2) \frac{A - \lambda_1 E}{\lambda_2 - \lambda_1}$$

$$\frac{A - \lambda_2 E}{\lambda_1 - \lambda_2} = \frac{1}{4-5} \begin{bmatrix} -5 & 1 \\ -20 & 4 \end{bmatrix} = \begin{bmatrix} 5 & -1 \\ 20 & -4 \end{bmatrix}; \quad \frac{A - \lambda_1 E}{\lambda_2 - \lambda_1} = \frac{1}{5-4} \begin{bmatrix} -4 & 1 \\ -20 & 5 \end{bmatrix} = \begin{bmatrix} -4 & 1 \\ -20 & 5 \end{bmatrix}$$

$$e^{At} = e^{\lambda_1 t} \frac{A - \lambda_2 E}{\lambda_1 - \lambda_2} + e^{\lambda_2 t} \frac{A - \lambda_1 E}{\lambda_2 - \lambda_1} = e^{4t} \begin{bmatrix} 5 & -1 \\ 20 & -4 \end{bmatrix} + e^{5t} \begin{bmatrix} -4 & 1 \\ -20 & 5 \end{bmatrix} = \begin{bmatrix} 5e^{4t} - 4e^{5t} & -e^{4t} + e^{5t} \\ 20e^{4t} - 20e^{5t} & -4e^{4t} + 5e^{5t} \end{bmatrix}$$

$$e^{At}x_0 = \begin{bmatrix} 5e^{4t} - 4e^{5t} & -e^{4t} + e^{5t} \\ 20e^{4t} - 20e^{5t} & -4e^{4t} + 5e^{5t} \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 15e^{4t} - 11e^{5t} \\ 60e^{4t} - 55e^{5t} \end{bmatrix}$$

$$A^{-1}(e^{At} - E) = \frac{e^{4t} - 1}{\lambda_1 - \lambda_2} \frac{A - \lambda_2 E}{\lambda_1 - \lambda_2} + \frac{e^{5t} - 1}{\lambda_2 - \lambda_1} \frac{A - \lambda_1 E}{\lambda_2 - \lambda_1} = \frac{e^{4t} - 1}{4} \begin{bmatrix} 5 & -1 \\ 20 & -4 \end{bmatrix} + \frac{e^{5t} - 1}{5} \begin{bmatrix} -4 & 1 \\ -20 & 5 \end{bmatrix} =$$

$$= \begin{bmatrix} \frac{25e^{4t} - 16e^{5t} - 9}{20} & \frac{-5e^{4t} + 4e^{5t} + 1}{20} \\ \frac{5e^{4t} - 4e^{5t} - 1}{20} & \frac{-e^{4t} + e^{5t}}{20} \end{bmatrix} \begin{bmatrix} 0 \\ 5 \end{bmatrix} = \begin{bmatrix} \frac{-5e^{4t} + 4e^{5t} + 1}{4} \\ -5e^{4t} + 5e^{5t} \end{bmatrix}$$

$$x_t = e^{At}x_0 + A^{-1}(e^{At} - E)b = \begin{bmatrix} 15e^{4t} - 11e^{5t} \\ 60e^{4t} - 55e^{5t} \end{bmatrix} + \begin{bmatrix} \frac{-5e^{4t} + 4e^{5t} + 1}{4} \\ -5e^{4t} + 5e^{5t} \end{bmatrix} = \begin{bmatrix} 55e^{4t} - 40e^{5t} + 1 \\ 55e^{4t} - 50e^{5t} \end{bmatrix}$$

Проверка:

$$1) x_t = \begin{bmatrix} y_t \\ z_t \end{bmatrix}; \quad y_t = \frac{55e^{4t} - 40e^{5t} + 1}{4}; \quad y_t' = 55e^{4t} - 50e^{5t} = z_t$$

$$2) x_0 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}; \quad y_0 = \frac{55 - 40 + 1}{4} = 4; \quad z_0 = 55 - 50 = 5$$